Huawei Access Node Routing Protocol Flap Resolution Procedure for BAS-ACC-37 at Basavanagudi POP

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5855

Description

The failure domain of Routing Protocol Flap, as denoted by alarm code ALARM_5855 with a severity of Critical on the Huawei access node BAS-ACC-37 situated at the Basavanagudi POP, stems from the intricate dynamics of routing protocols such as BGP, OSPF, and ISIS. These protocols are fundamental to the establishment and maintenance of network connectivity, relying on the exchange of routing information between nodes to construct and update routing tables. The specific role of an access node, acting as the ingress point for subscriber traffic into the network, amplifies the impact of routing instability. Flaps or churns in BGP routes, loss of OSPF adjacencies, or ISIS LSP flooding can lead to transient or persistent network outages, as the network struggles to converge on a stable topology. This SOP is designed to methodically diagnose and rectify the root causes of such routing protocol flaps on Huawei equipment, leveraging the vendor's CLI dialect for precision and effectiveness.

Pre-requisite Checks

- Dispatch remote hands to Basavanagudi POP for on-site support
- Verify redundant link status to ensure no single point of failure
- Confirm power supply and environmental conditions are within operational specifications

Potential Root Causes

Cause	Probability
Misconfigured BGP attributes leading to route churn	High
Physical layer issues causing OSPF adjacency loss	Medium
ISIS LSP flooding due to network topology changes	Low

Diagnostic Commands & Expected Outputs

```
Command: display ip bgp peer, Expected Output: BGP peer information, including state and last error, Description: This command isolates issues in the BGP control plane by displaying the state of BGP peers, which can indicate if there are any problems with the establishment or maintenance of BGP sessions.

Command: display ospf peer, Expected Output: OSPF neighbor information, including state and dead timer, Description: By examining the state of OSPF neighbors, this command helps diagnose issues related to the formation and stability of OSPF adjacencies, crucial
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```
for intra-area routing.

Command: display isis peer, Expected Output: ISIS neighbor
information, including state and last error, Description: This command
provides insights into the ISIS control plane, enabling the diagnosis
of issues related to the establishment and maintenance of ISIS
adjacencies, which are critical for routing within ISIS areas.
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Resolution Steps

Step 1: Step: Adjust BGP Attributes for Stability, Command: bgp peer ebgp-max-retain , Description: Modify BGP attributes to mitigate route churn by adjusting the maximum number of routes retained, which can help stabilize BGP sessions and reduce the likelihood of route flaps.

Step 2: Step: Rectify OSPF Adjacency Issues, Command: ospf area network , Description: Reconfigure OSPF areas or network types to ensure proper adjacency formation, addressing issues related to OSPF neighbor discovery and database synchronization.

Step 3: Step: Configure ISIS to Prevent LSP Flooding, Command: isis enable level-1, Description: Enable ISIS level-1 routing to limit the scope of LSP flooding, ensuring that topology changes do not excessively impact network stability by controlling the dissemination of LSPs.

Post-Check Verification

```
Command: display ip routing-table, Expected Output: Current IP routing
table, Description: Verify that the routing table reflects the
expected topology and that routes are being properly advertised and
learned.

Command: display bgp routing-table, Expected Output: Current BGP
routing table, Description: Confirm that BGP routes are stable and
correctly propagated throughout the network.

Command: display ospf routing-table, Expected Output: Current OSPF
routing table, Description: Check that OSPF routes are correctly
calculated and reflect the network topology.
```

Roll-back Procedures

- Step: Revert BGP Configuration, Command: undo bgp peer ebgp-max-retain, Description: If adjustments to BGP attributes do not resolve the issue or introduce new problems, revert these changes to their original state to maintain consistency with the pre-intervention configuration.

- Step: Restore OSPF Configuration, Command: undo ospf area network , Description: If modifications to OSPF configurations do not yield the desired outcome, undo these changes to revert to the previous state, ensuring that any new issues introduced by the changes are mitigated.

- Step: Disable ISIS Changes, Command: undo isis enable level-1, Description: If enabling ISIS level-1 routing does not prevent LSP flooding as intended, or if it introduces unforeseen issues, disable this feature to return to the original ISIS configuration.